

CODEFEST 8

SNN

$$N = 1024$$

$$f = 50 \text{ Hz}$$

$$1) R = N \times f$$

$$= 1024 \times 50$$

$$= 51,200 \text{ Hz spikes/sec}$$

=

$$2) \text{AER packet} = 10 \text{ bit - address}$$

$$6 \text{ bit - timestamp}$$

$$4 \text{ bit - framing/parity}$$

$$20 \text{ - bit / packet}$$

$$B = R \times 20$$

$$= 51,200 \times 20$$

$$= 1,024,000 \text{ bits/sec}$$

$$\approx 1 \text{ Mbit/sec}$$

3) Interconnection

SPI

B.W

< 50 Mbits/s

Comparison to above BW

~~Not~~ possible

I2C

< 3.4 Mbits/s

possible

AXI-Lite

100 Mbits/s

possible

All interfaces can sustain the mean rate.

The lowest-complexity interface that suffices is I2C.

4) Burst:

25% of 1024 neurons in 1ms window

$$\frac{25 \times 1024}{100} = 256 \text{ neurons fire in 1ms window}$$

$$\begin{aligned}
 \text{peak BW} &= R \times 20 \\
 &= 256 \times 10^3 \times 20 \\
 &= 5,120,000 \text{ ~~spikes~~ bits/sec.} \\
 &\approx 5 \text{ Mbits/sec}
 \end{aligned}$$

$$\begin{aligned}
 \text{Burst to Mean Ratio} &= \frac{\text{Peak BW}}{\text{Mean B.W.}} \\
 &= \frac{5 \text{ Mbits/sec}}{1 \text{ Mbit/sec.}} \\
 &= 5
 \end{aligned}$$

The I²C interface chosen cannot sustain the burst rate. Buffering is required.

$$\begin{aligned}
 \text{Overflow in 1ms.} &= \text{peak BW} - \text{I}^2\text{C BW} = 256 \times 20 - 3,400,000 \times 0.001 \\
 &= 5120 - 3400 \\
 &= 1720 \text{ bits.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Overflow in terms of packets} \\
 &= \frac{1720}{20} = 86 \text{ packets.}
 \end{aligned}$$

So, roughly the size of the buffer = 256 bytes.

$$\begin{aligned}
 5) \text{ non-AEK BW} &= 1024 \times 10^3 \\
 &\approx 1 \text{ Mbits/sec.}
 \end{aligned}$$

$$\begin{aligned}
 \text{AEK to frame BW ratio at mean firing rate} \\
 &= \frac{1 \text{ Mbits/sec}}{1 \text{ Mbits/sec}} = 1
 \end{aligned}$$

Crossover firing rate.

$$B_{AER} = B_{frame}$$
$$N \times f \times (\text{bits/packet}) = \frac{N \times}{T_{frame}}$$

$$f = \frac{1}{10^{-3}} \times \frac{1}{\text{bits/packet}}$$
$$= \frac{1}{10^{-3}} \times \frac{1}{20}$$
$$= 50 \text{ Hz}$$

AER is the right choice when neurons fire below crossover frequency rather than continuously sampling the neurons as the packet overhead otherwise would dominate.